Phase 2: Information Design & IA Mapping

1. Empirical Card Sorting Study

1.1 Method

I ran an open card sort in Lucidchart with three participants. Each participant dragged the same 16 cards from an unsorted deck into category frames and could rename the frames. The 16 cards come from the elective lookup in Phase 1: course content, filters and actions.

Number	Card
1	Course Prefix & Number
2	Course Title
3	Course Description / Syllabus Overview
4	Prerequisites Checklist
5	Term / Semester Availability
6	Section CRN / Call Number
7	Instructor Name & Contact Info
8	Real-Time Seat Availability
9	Class Meeting Days & Times
10	Class Location / Modality
11	Filter by Semester / Term
12	Filter by Course Status (Open Seats Only)
13	Side-by-Side Course Comparison Tool
14	Add to Personal Schedule Builder
15	Degree Requirement Progress Tracker
16	Global Navigation Back to Degree Audit

1.2 What each participant did

Participant 1:

Category label chosen	Cards placed
Course Details (7)	Prerequisites Checklist; Course Title; Course Description / Syllabus Overview; Course Prefix & Number; Term / Semester Availability; Section CRN / Call Number; Real-Time Seat Availability
Filter Tools (2)	Filter by Semester / Term; Filter by Course Status (Open Seats Only)
Schedule Info (3)	Instructor Name & Contact Info; Class Location / Modality; Class Meeting Days & Times
General Information (4)	Global Navigation Back to Degree Audit; Add to Personal Schedule Builder; Degree Requirement Progress Tracker; Side-by-Side Course Comparison Tool

Participant 2:

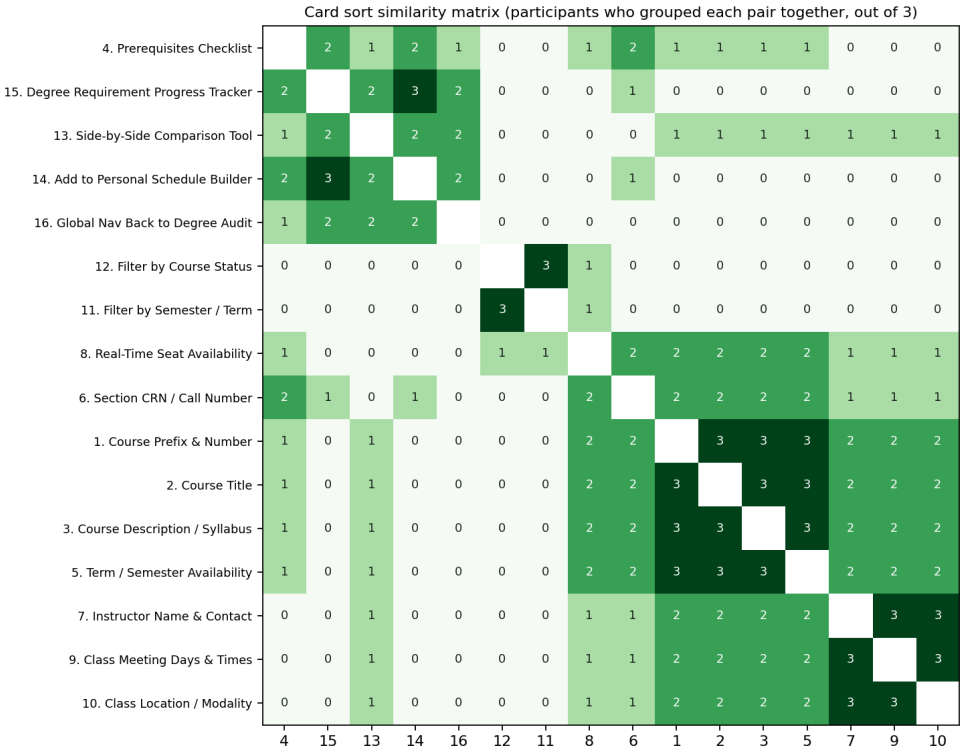
Category label chosen	Cards placed
Filter Options (3)	Filter by Course Status (Open Seats Only); Real-Time Seat Availability; Filter by Semester / Term
Degree Progress (4)	Prerequisites Checklist; Degree Requirement Progress Tracker; Add to Personal Schedule Builder; Section CRN / Call Number
Course Details (8)	Course Description / Syllabus Overview; Course Title; Class Location / Modality; Instructor Name & Contact Info; Side-by-Side Course Comparison Tool; Class Meeting Days

	& Times; Course Prefix & Number; Term / Semester Availability
Actions(1)	Global Navigation Back to Degree Audit

Participant 3:

Category label chosen	**Cards placed**
My Degree / Actions (5)	Prerequisites Checklist; Side-by-Side Course Comparison Tool; Add to Personal Schedule Builder; Degree Requirement Progress Tracker; Global Navigation Back to Degree Audit
Course Details (9)	Real-Time Seat Availability; Course Description / Syllabus Overview; Course Title; Course Prefix & Number; Term / Semester Availability; Class Location / Modality; Class Meeting Days & Times; Section CRN / Call Number; Instructor Name & Contact Info
Filters & Controls (2)	Filter by Course Status (Open Seats Only); Filter by Semester / Term

1.3 Similarity matrix



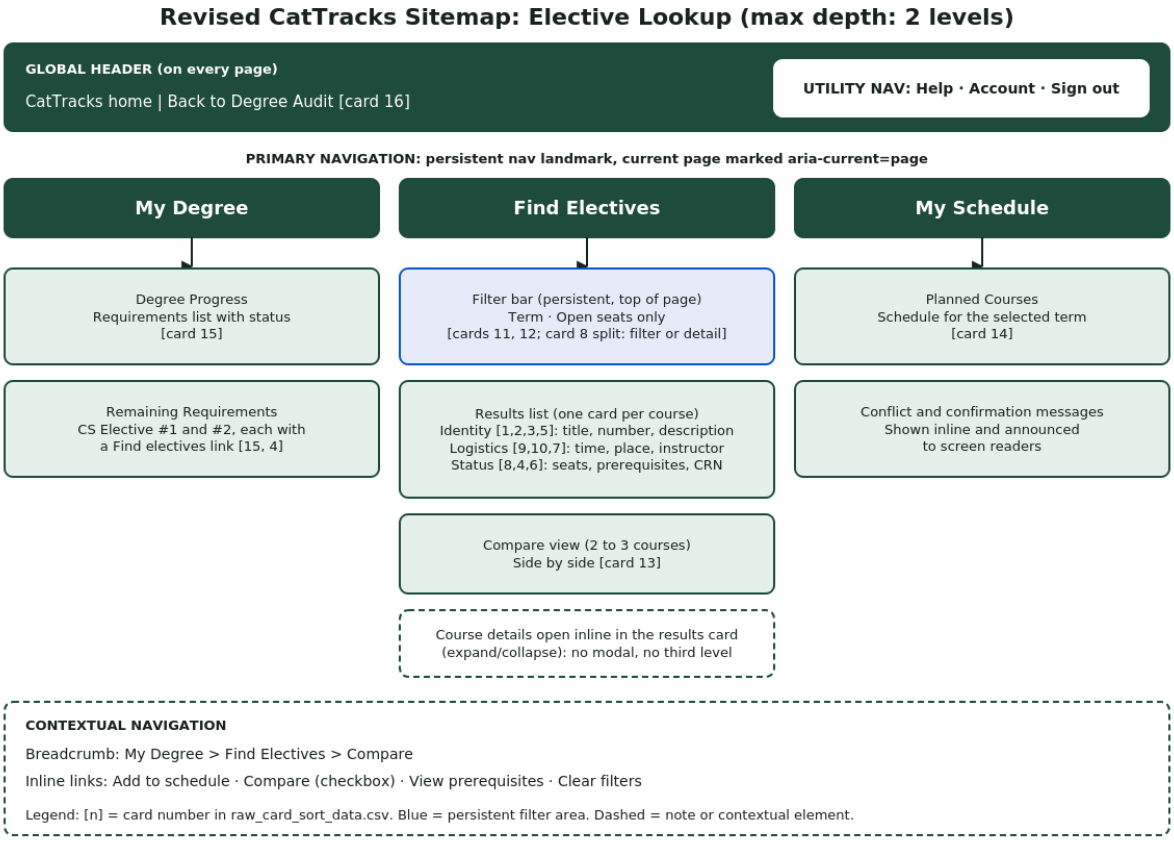
1.4 Where participants disagreed

Card	Participant 1	Participant 2	Participant 3
Real-Time Seat Availability	Course Details	Filter Option	Course Details
Side-by-Side Course Comparison Tool	General Information	Course Details	My Degree / Actions
Prerequisites Checklist	Course Details	Degree Progress	My Degree / Actions
Section CRN / Call Number	Course Details	Degree Progress	Course Details
Global Navigation Back to Degree Audit	General Information	Actions	My Degree / Actions

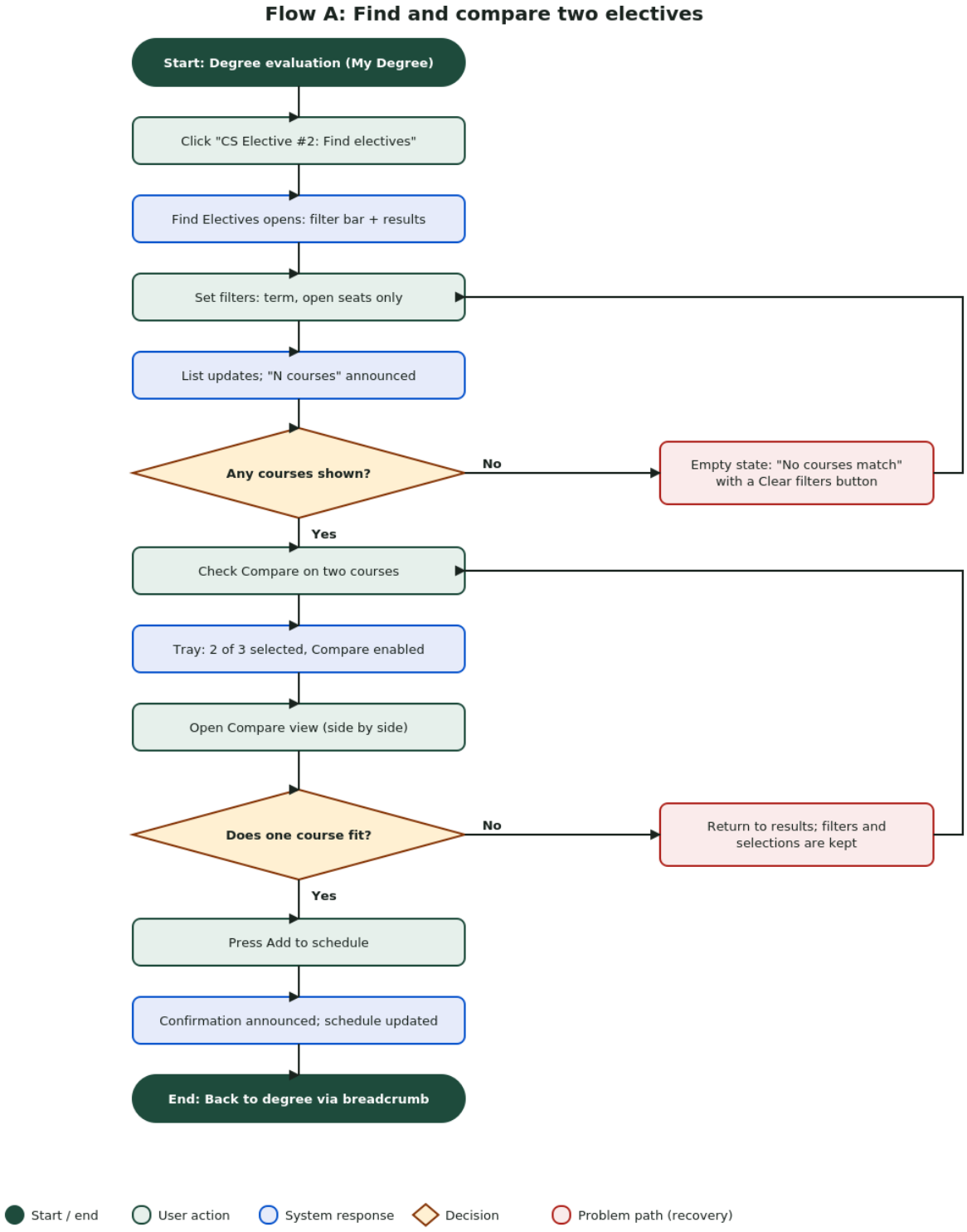
1.5 Patterns from Participants

- All three kept filters separate from course details, so filtering is its own mental category.
- Seat availability: one participant treated it as a filter, two treated it as course information.Two participants treated Compare as an action, not as course information.
- Prerequisites went with degree progress for two participants, which suggests students see them as an eligibility check.
- The Global Navigation card never joined course content, so it belongs in the header.
- Section CRN had no consensus (three different placements), so I treated it as secondary detail.

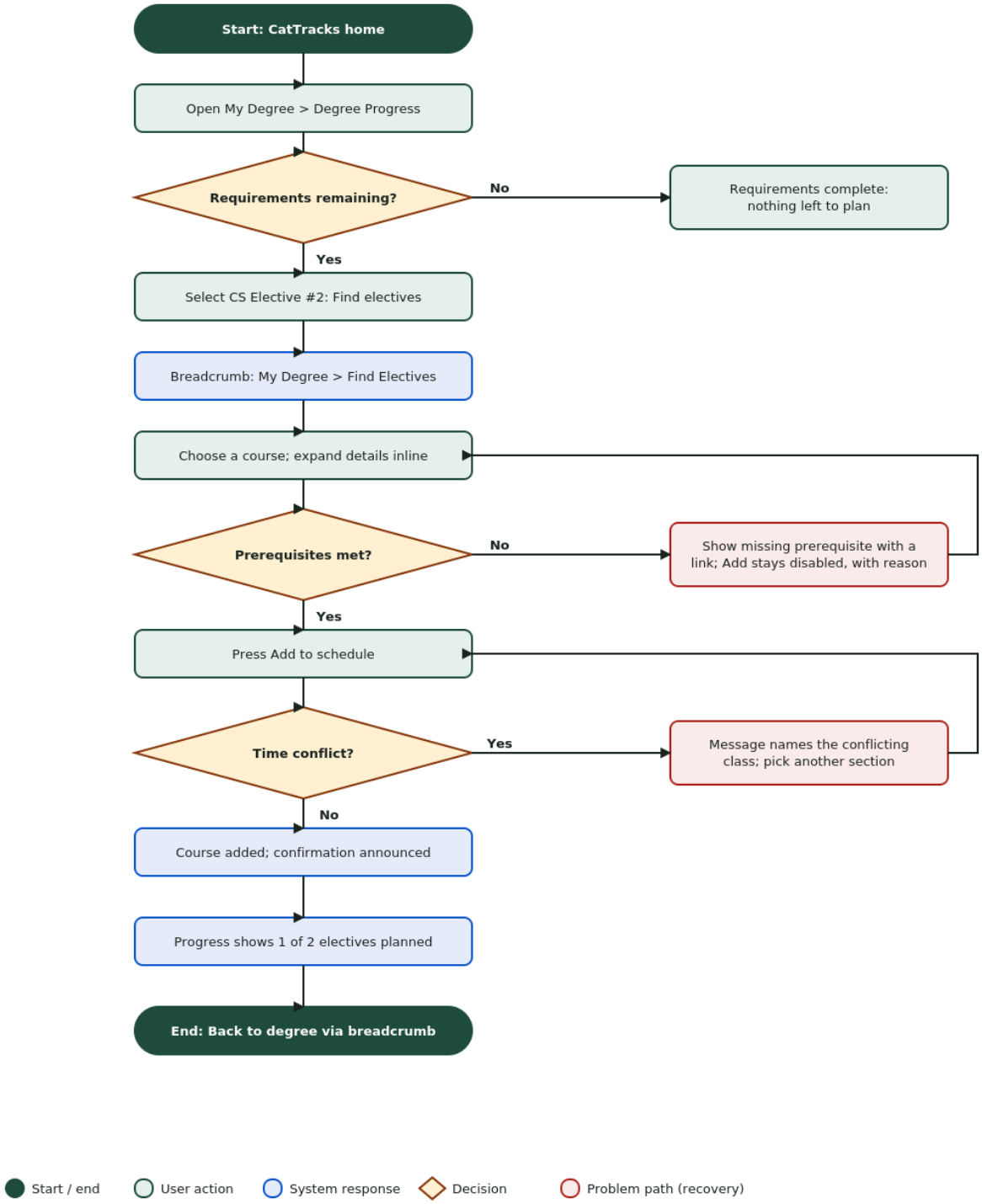
2. Revised Sitemap and IA Blueprint



3. User Flows


Flow A: Find and compare two electives
Start: Degree evaluation (My Degree)
Click "CS Elective #2: Find electives"
Find Electives opens: filter bar + results
Set filters: term, open seats only
List updates; "N courses" announced
Any courses shown?
No
Empty state: "No courses match" with a Clear filters button
Yes
Check Compare on two courses
Tray: 2 of 3 selected, Compare enabled
Open Compare view (side by side)
Does one course fit?
No
Return to results; filters and selections are kept
Yes
Press Add to schedule
Confirmation announced; schedule updated
End: Back to degree via breadcrumb
Start / end
User action
System response
Decision
Problem path (recovery)

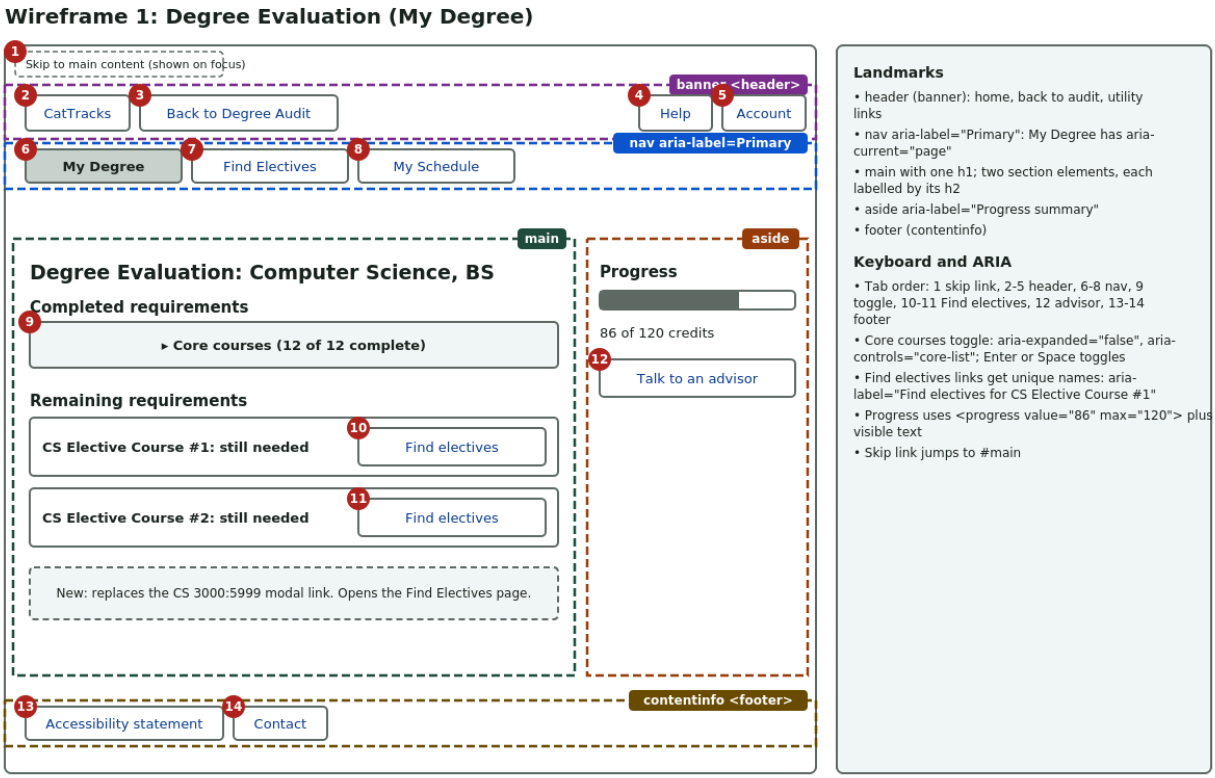
Flow B: Check requirements, then add an elective



4. Wireframes with Semantic and Accessibility Annotations

4 low-fidelity wireframes cover every screen in both flows. Red numbered badges show the keyboard tab order. Colored dashed outlines show landmarks. Placeholder text and sample course data are for layout only. The full HTML skeletons are in IA_Mapping/semantic_annotations/semantic_html_annotations.txt

Wireframe 1: Degree Evaluation



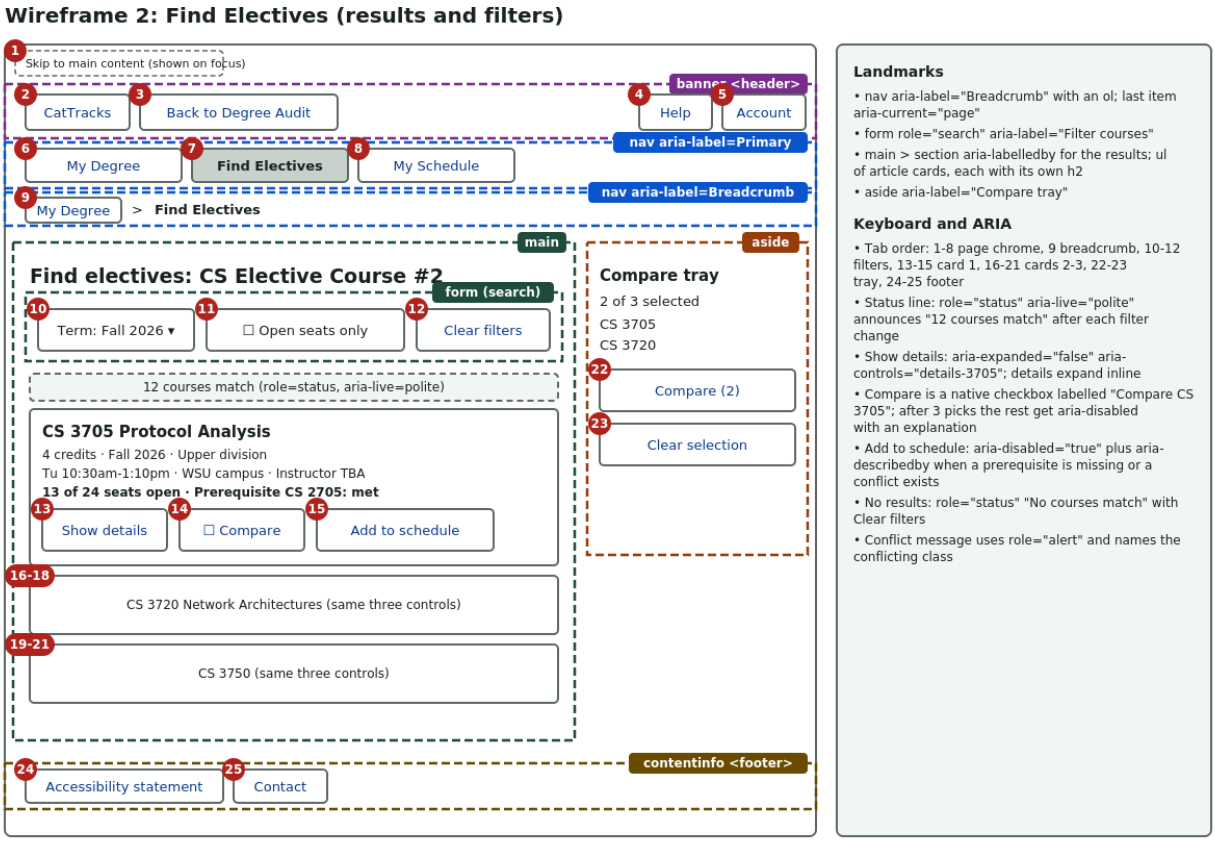
Landmarks

- header (banner): home, back to audit, utility links
- nav aria-label="Primary": My Degree has aria-current="page"
- main with one h1; two section elements, each labelled by its h2
- aside aria-label="Progress summary"
- footer (contentinfo)

Keyboard and ARIA

- Core courses toggle: aria-expanded="false", aria-controls="core-list"; Enter or Space toggles
- Find electives links get unique names: aria-label="Find electives for CS Elective Course #1"
- Progress uses <progress value="86" max="120"> plus visible text
- Skip link jumps to #main

Wireframe 2:



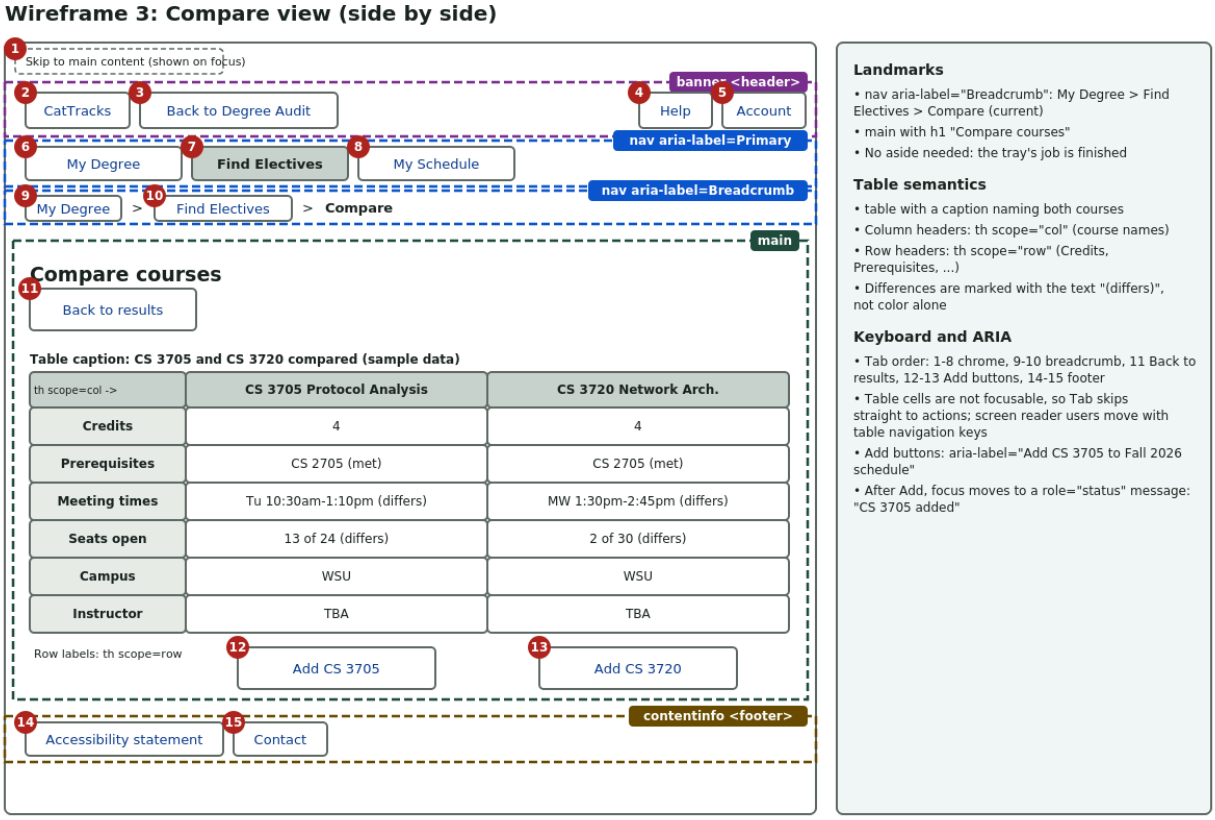
Landmarks

- form role="search" aria-label="Filter courses"
- main > section aria-labelledby for the results; ul of article cards, each with its own h2
- aside aria-label="Compare tray"

Keyboard and ARIA

- Status line: role="status" aria-live="polite" announces "12 courses match" after each filter change
- Show details: aria-expanded="false" aria-controls="details-3705"; details expand inline
- Compare is a native checkbox labelled "Compare CS 3705"; after 3 picks the rest get aria-disabled with an explanation
- Add to schedule: aria-disabled="true" plus aria-describedby when a prerequisite is missing or a conflict exists
- No results: role="status" "No courses match" with Clear filters
- Conflict message uses role="alert" and names the conflicting class

Wireframe 3:



Landmarks

- main with h1 "Compare courses"
- No aside needed: the tray's job is finished

Table semantics

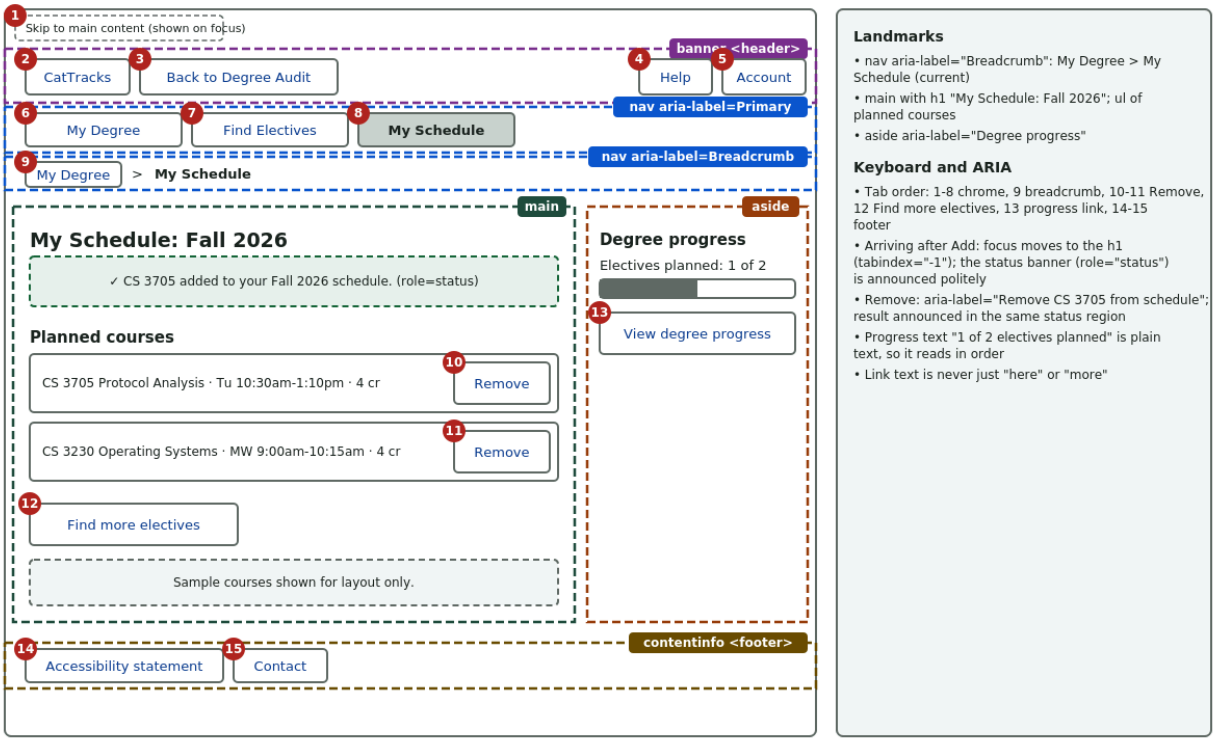
- table with a caption naming both courses
- Column headers: th scope="col" (course names)
- Row headers: th scope="row" (Credits, Prerequisites, ...)
- Differences are marked with the text "(differs)", not color alone

Keyboard and ARIA

- Table cells are not focusable, so Tab skips straight to actions; screen reader users move with table navigation keys
- Add buttons: aria-label="Add CS 3705 to Fall 2026 schedule"
- After Add, focus moves to a role="status" message: "CS 3705 added"

Wireframe 4:

Wireframe 4: My Schedule (confirmation)



Landmarks

- main with h1 "My Schedule: Fall 2026"; ul of planned courses
- aside aria-label="Degree progress"

Keyboard and ARIA

- Arriving after Add: focus moves to the h1 (tabindex="-1"); the status banner (role="status") is announced politely
- Remove: aria-label="Remove CS 3705 from schedule"; result announced in the same status region
- Progress text "1 of 2 electives planned" is plain text, so it reads in order
- Link text is never just "here" or "more"

5. Post-Mortem Reflection

How did the empirical data from your card sorting experiment challenge your initial assumptions about how the system's navigation should be structured?

After Phase 1, I thought the fix was making a better organized list, remove unavailable courses, and add comparison tools but the card sorting changed my assumption in three ways. First, all three participants separated filters from course details, so filters needed to be their own persistent control, not a feature buried inside a list. Second, participants split course content into identity (title,

number, description) and logistics (time, place, instructor), so I stopped treating a course as one block and ordered each result card in that way. Third, seat availability split between filters and details, and prerequisites went with degree progress for two participants. That told me students use these as decision tools, so they now appear on the result card and explain disabled buttons. The sort also kept Global Navigation away from course content, which moved Back to Degree Audit into the header. Without the data I would have designed from my own mental model, which is the bias Phase 1 warned about.

Describe how your annotated ARIA landmarks and keyboard tab sequences ensure that screen reader users navigate your revised information architecture as efficiently as sighted users.

A sighted user scans the page and jumps to the filters or results, screen reader users also need equivalent shortcuts. The shared layout provides them with a skip link that bypasses the header, labelled landmarks to let a user jump straight to the filters or the compare tray, and a single h1, h2 headings give a scannable outline. Tab order follows the visual order, so the keyboard path matches what sighted users see, and I avoided positive tabindex values that would break it. Dynamic changes are announced by a polite status region that reports "12 courses match" after filtering, and the schedule confirmation is announced without moving focus away from the task. The Compare table uses real header cells, so a user hears the course and the attribute at each cell instead of having to remember the columns, which was the Phase 1 recall problem. Disabled Add buttons use aria-disabled with a described reason, so they stay discoverable.